

PinCity: Multiplayer AR system with Shape-Changing Displays for Dynamic Urban Design

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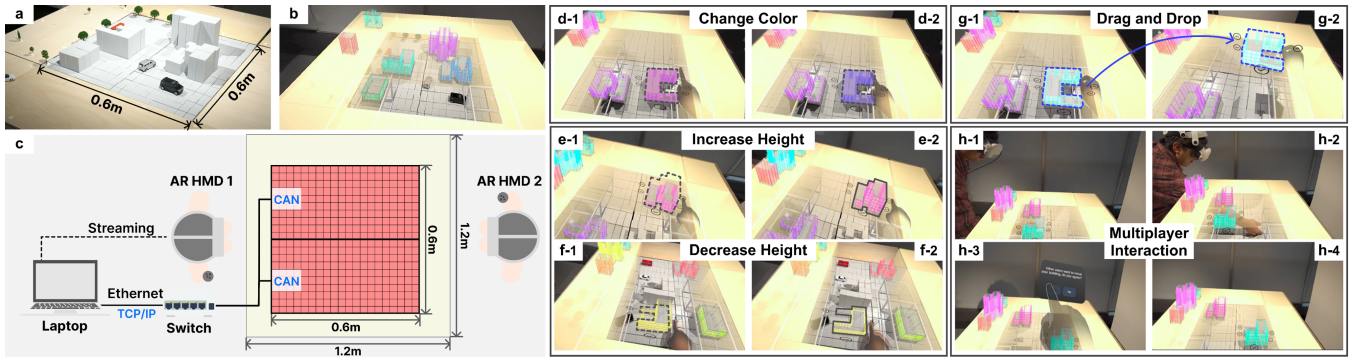


Figure 1: Overview of the PinCity system. (a) Physical setup of the city scene on the pin-array-based shape-changing display. (b) Virtual city models in AR. (c) System architecture, showing the communication flow between the laptop, AR HMDs, and the shape-changing display. (d) Color change interaction. (e) Increasing the height of buildings. (f) Decreasing the height of buildings. (g) Drag-and-drop function for moving objects within the design space. (h) Multiplayer interaction, where two users collaborate in real-time to modify the design, with synchronized changes reflected in both AR and shape-changing display.

Abstract

Urban design, facilitated by the integration of emerging technologies such as AR, has increasingly transitioned into an interactive and collaborative process. This paper presents PinCity, an AR urban design system that enables multiple users to have real-time tangible interaction between virtual urban models and a physical pin-array-based shape-changing display within a synchronized augmented environment. By merging a virtual urban scene with physical feedback and enabling multi-user collaboration, PinCity facilitates an intuitive experience co-design.

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CCS Concepts

• Human-centered computing → Displays and imagers; Visualization systems and tools.

Keywords

Pin-array, Shape changing Display, Visualization

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1 Introduction

In recent years, urban design has developed into an interactive process. By integrating physical city scenes with emerging visual technologies such as Virtual Reality (VR) and Augmented Reality (AR), users can adjust scene layouts and virtual objects, thus offering more possibilities for future urban design. Especially, AR system

delivered through Head-Mounted Displays (HMDs) and projection mapping provide more balanced interaction methods between physical environments and virtual objects, enhancing seamless discourse between the real world and the virtual design space [Astriani et al. 2022; Ihara et al. 2023; Lock et al. 2019].

Previous research has primarily focused on combining AR with static physical interfaces, which support visual modifications yet without corresponding physical changes [Alonso et al. 2018; Leithinger et al. 2014; Noyman et al. 2019; Piper et al. 2002]. To achieve more dynamic urban designs, researchers are gradually transforming static physical urban scenes into tangible interfaces, enabling people to modify and iterate the urban design more intuitively. On the other hand, there is research that utilizes a shape-changing display to generate environments, such as terrain [Gomi et al. 2024; Je et al. 2021; Suzuki et al. 2021; Teng et al. 2019] and city views [Follmer et al. 2013; Ishii et al. 2015; Leithinger et al. 2013; MIT Media Lab City Science [n. d.]]. However, neither approach explored AR using HMDs with shape-changing displays for multi-user urban design.

In this paper, we introduce PinCity, an AR urban design system that supports shape-changing interfaces and multiplayer interactions. PinCity integrates a pin-array-based shape-changing display that supports synchronized and tangible shape changes in response to modifications of virtual objects. Users can place virtual objects and modify their properties in the AR environment, while also interacting with real-time, tangible changes on the shape-changing display. Furthermore, PinCity supports co-design among multiple users, allowing them to collaborate more intuitively in designing and inspiring augmented environments.

2 System Design

Our PinCity system integrates a physical pin table with an AR application to support dynamic, co-located design interactions.

The pin table is a 1200 mm × 1200 mm pin-based shape-changing display that comprises 400 individually actuated pins arranged in a 20 × 20 grid, covering a movable area of 600 mm × 600 mm. Each pin is a cuboid measuring 30 mm × 30 mm × 367.5 mm and is capable of vertical movement within a range of 0 to 300 mm. This allows the display to form a variety of 3D shapes in real-time dynamically.

The AR application is developed in Unity 6000.0.51f1 and runs on a PC system with an Intel Core i9 processor and an NVIDIA GeForce RTX 4070 GPU. The application includes six different virtual building models, each with adjustable height and color. Users can manipulate these models through the AR interface, modifying their position and properties.

Once the user makes a modification, the AR system sends the updated shape data to the host laptop. The laptop then generates a corresponding JSON file and transmits it via Ethernet using the TCP/IP protocol to the pin table's IP address. The pin table processes the JSON file and distributes the data to each pin via the CAN bus. Each pin then renders the corresponding physical shape, ensuring synchronization between the virtual and physical representations.

3 Interaction

Our system enables multiple participants to explore the virtual urban design and interact with the physical interface together. The

details of the interaction are shown in Figure 1. Participants can place virtual buildings onto the physical pin table and manipulate them using their hands. Touching a building changes its color, while pressing the corresponding button adjusts its height. Once a user finalizes their edits, they can initiate synchronization with the physical process by pressing the green "Run" button located in the bottom right corner of the interface. This triggers a pop-up notification that appears in the other user's view. The second user has the option to accept or reject the changes. If accepted, the virtual city scene is synchronized across both headsets, and the physical pin table adjusts to reflect the updated design. Users can utilize PinCity as a platform for engaging in urban planning, visualization, and discussion, promoting a more interactive and collaborative approach to designing urban environments in AR.

4 Conclusion

In this paper, we introduce PinCity, an AR urban design system that combines dynamic tangible interfaces with AR to facilitate collaborative co-creation. PinCity integrates a pin-array-based, shape-changing display, enabling multiple users to intuitively co-create by modifying virtual objects alongside their corresponding physical shapes in real-time. We envision PinCity as a future platform for urban design, promoting a collaborative approach to visualize and shape urban environments through seamless AR and tangible interactions. In the future, we plan to explore how our system can improve remote collaboration in co-design scenarios and integrate cross-reality design workflows to make the collaborative design process more effective.

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